

AI FOR HUMANITARIAN TECHNOLOGY

PROBLEM STATEMENT ID	PROBLEM STATEMENT	PROBLEM STATEMENT DESCRIPTION
AI-01	Hyperlocal AI Disaster Early-Warning & Multilingual Alert System	Develop an AI-based Early Warning System that uses IMD APIs, satellite data, NDMA datasets, social media signals, and simulated IoT sensor data to predict floods, cyclones, and landslides 12–72 hours in advance. The system displays risk levels and confidence scores on a Streamlit/React dashboard and sends multilingual voice and SMS alerts through Twilio/WhatsApp to communities with limited internet access, providing localized, timely disaster warnings using simulated sensor feeds without requiring physical IoT hardware.
AI-02	AI Health Screener for Rural Primary Health Centre Workers	Develop an AI-based Health Screening Assistant for PHC workers (ASHA/ANMs) that collects patient symptoms (via voice or form), vital signs, and medical history to generate low/medium/high risk stratification, referral guidance, and chronic disease alerts using an ML classifier trained on ICMR/NHP datasets. The mobile-first web app supports English and one Indian regional language (Kannada/Hindi), includes a basic patient tracker, integrates Google TTS/IndicTTS for voice assistance, and is designed for future compatibility with ABDM sandbox APIs.

AI-03	Affordable AI-Powered AAC Tool for Non-Verbal Differently-Abled Individuals	Develop an AI-powered Augmentative and Alternative Communication (AAC) tool for non-verbal individuals that enables communication through webcam-based eye-gaze tracking (MediaPipe Iris), facial gestures, or a symbol board with text-to-speech. The system adapts to user behavior by learning frequently used patterns, supports Hindi and one regional language, and provides offline, affordable communication assistance using technologies like MediaPipe, OpenCV, Coqui TTS, and SQLite.
AI-04	WhatsApp/IVR Crop Disease & Soil Advisory for Marginal Farmers	Develop an AI-powered Crop Advisory System accessible via WhatsApp or IVR that accepts crop photos or voice descriptions to detect diseases and pests using a CNN (PlantVillage dataset), provides farmer-language advisory (Hindi + Kannada), offers soil health recommendations based on NPK/pH values, and suggests relevant government schemes such as PM-KISAN and the Soil Health Card, ensuring easy access for small and marginal farmers with basic phones.
AI-05	Real-Time Indian Sign Language Recognition & Translation System	Develop a real-time Indian Sign Language (ISL) Recognition System that uses a webcam and MediaPipe Holistic to detect hand and facial landmarks, recognizes 50+ ISL signs using LSTM/Transformer models, and converts them into text

		and optional speech (TTS). The system also includes few-shot learning to add new signs without full retraining, achieving 80%+ accuracy for use in education, healthcare, and government services.
AI-06	Federated AI Platform for Privacy-Preserving Disease Surveillance Across PHCs	Develop a Federated Learning-Based Disease Outbreak Detection System where simulated PHC nodes train local models on symptom data without sharing raw patient records. A FedAvg aggregator combines these models to detect disease outbreaks and anomalous symptom clusters, with optional differential privacy for enhanced data security. The system displays a district-level disease risk heatmap, ensuring DPDP Act 2023 compliance while enabling real-time, privacy-preserving outbreak monitoring.